Chapter 5: Designing IP addressing and Selecting Routing Protocol

Designing an IP Addressing Plan

Private and Public IPv4 Addresses

The IP address space is divided into public and private spaces. Private addresses are reserved IP addresses that are to be used only internally within a company's network, not on the Internet. Private addresses must therefore be mapped to a company's external registered address when sending anything on the Internet. Public IP addresses are provided for external communication. Figure 6-1illustrates the use of private and public addresses in a network.

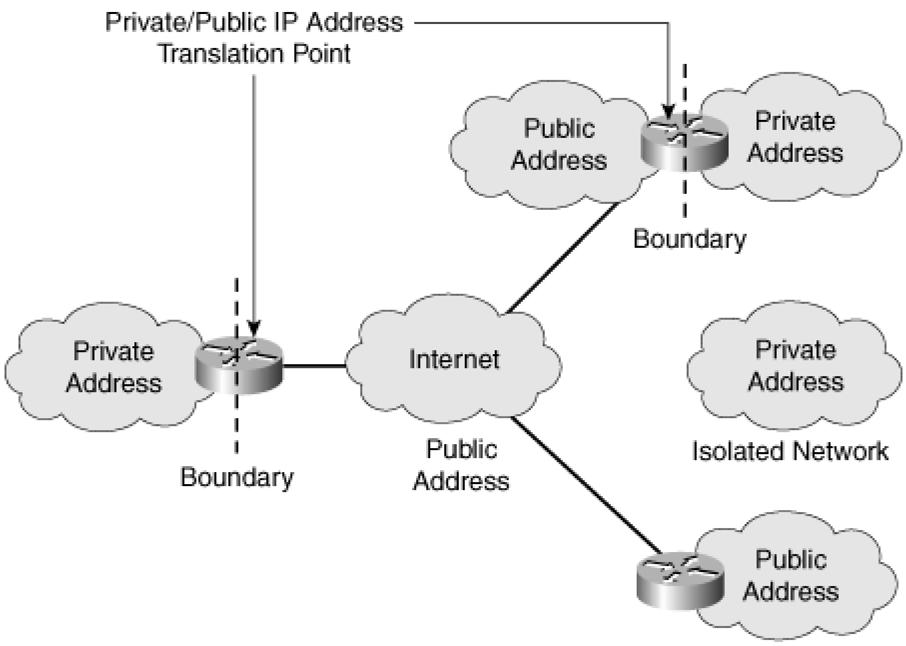

Figure 6-1: Private and Public Addresses Can Be Used in a Network

RFC 1918, *Address Allocation for Private Internets*, defines the private IP addresses as follows:

- 10.0.0.0 to 10.255.255.255
- 172.16.0.0 to 172.31.255.255
- 192.168.0.0 to 192.168.255.255

The remaining addresses are public addresses.

Private versus Public Address Selection Criteria

Very few public IP addresses are currently available, so Internet service providers (ISPs) can assign only a subset of Class C addresses to their customers. Therefore, in most cases, the number of public IP addresses assigned to an organization is inadequate for addressing their entire network.

The solution to this problem is to use private IP addresses within a network and to translate these private addresses to public addresses when Internet connectivity is required. When selecting addresses, the network designer should consider the following questions:

- Are private, public, or both IP address types required?
- How many end systems need only access to the public network? This is the number of end systems that need a limited set of external services (such as e-mail, file transfer, or web browsing) but do not need unrestricted external access. These end systems do not have to be visible to the public network.
- How many end systems must have access to and be visible to the public network? This is the number of Internet connections and various servers that must be visible to the public network (such as public servers and servers used for e-commerce, such as web servers, database servers, and application servers) and defines the number of required public IP addresses. These end systems require globally unambiguous IP addresses.
- Where will the boundaries between the private and public IP addresses be, and how will they be implemented?

Interconnecting Private and Public Addresses

According to its needs, an organization can use both public and private addresses. A router or firewall acts as the interface between the network's private and public sections.

When private addresses are used for addressing in a network and this network must be connected to the Internet, Network Address Translation (NAT) or Port Address Translation (PAT) must be used to translate from private to public addresses and vice versa. NAT or PAT is required if accessibility to the public Internet or public visibility is required.

Static NAT is a one-to-one mapping of an unregistered IP address to a registered IP address.

Dynamic NAT maps an unregistered IP address to a registered IP address from a group of registered IP addresses.

NAT overloading, or *PAT*, is a form of dynamic NAT that maps multiple unregistered IP addresses to a single registered IP address by using different port numbers. As shown in Figure 6-2, NAT or PAT can be used to translate the following:

- **One private address to one public address:** Used in cases when servers on the internal network with private IP addresses must be visible from the public network. The translation from the server's private IP address to the public IP address is defined statically.
- **Many private addresses to one public address:** Used for end systems that require access to the public network but do not have to be visible to the outside world.
- **Combination:** It is common to see a combination of the previous two techniques deployed throughout networks.

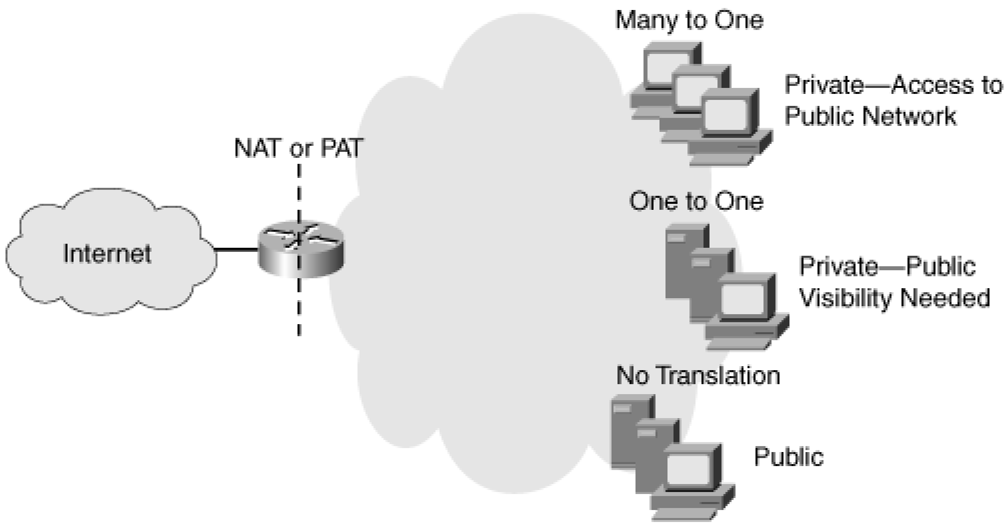

Figure 6-2: Private to Public Address Translation

Private IP addresses are used throughout the Enterprise Campus, Enterprise Branch, and Enterprise Teleworker modules. The following modules include public addresses:

- The Internet Connectivity module, where public IP addresses are used for Internet connections and publicly accessible servers.
- The E-commerce module, where public IP addresses are used for the database, application, and web servers.
- The Remote Access and virtual private network (VPN) module, the Enterprise Data Center module, and the WAN and metropolitan-area network (MAN) and Site-to-Site VPN module, where public IP addresses are used for certain connections.

Determining the Size of the Network

The first step in designing an IP addressing plan is determining the size of the network to establish how many IP subnets and how many IP addresses are needed on each subnet. To gather this information, answer the following questions:

- **How many locations does the network consist of?:** The designer must determine the number and type of locations.
- **How many devices in each location need addresses?:** The network designer must determine the number of devices that need to be addressed, including end systems, router interfaces, switches, firewall interfaces, and any other devices.
- **What are the IP addressing requirements for individual locations?:** The designer must collect information about which systems will use dynamic addressing, which will use static addresses, and which systems can use private instead of public addresses.
- **What subnet size is appropriate?:** Based on the collected information about the number of networks and planned switch deployment, the designer estimates the appropriate subnet size. For example, deploying 48-port switches would mean that subnets with 64 host addresses would be appropriate, assuming one device per port.

Determining the Network Topology

Initially, the designer should acquire a general picture of the network topology; this will help determine the correct information to gather about network size and its relation to the IP addressing plan.

With this general network topology information, the designer determines the number of locations, location types, and their correlations. For example, the network location information for the topology shown in Figure 6-4 is shown in Table 6-1.

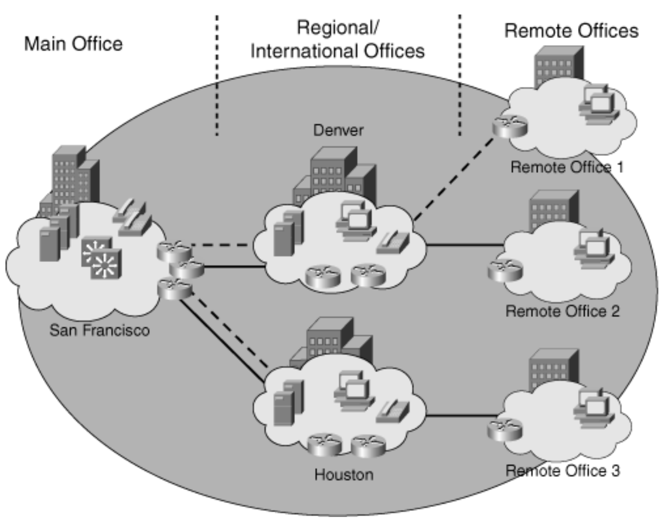

Figure 6-4: Sample Network Topology

Table 6-1: Network Location Information for the Topology in Figure 6-4

Location	Type	Comments
San Francisco	Main office	The central location where the majority of users are located
Denver	Regional office	Connects to the San Francisco main office
Houston	Regional office	Connects to the San Francisco main office
Remote Office 1	Remote office	Connects to the Denver regional office
Remote Office 2	Remote office	Connects to the Denver regional office
Remote Office 3	Remote office	Connects to the Houston regional office

Size of Individual Locations

The network size, in terms of the IP addressing plan, relates to the number of devices and interfaces that need an IP address. To establish the overall network size in a simplistic way, the designer determines the approximate number of workstations, servers, Cisco IP phones, router interfaces, switch management and Layer 3 interfaces, firewall interfaces, and other network devices at each location. This estimate provides the minimum overall number of IP addresses that are needed for the network. Table 6-2 provides the IP address requirements by location for the topology shown in Figure 6-4.

Table 6-2: IP Addressing Requirements by Location for the Topology in Figure 6-4

Location	Office Type	Workstations	Servers	IP Phones	Router Interfaces	Switches	Firewall and Other Device Interfaces	Reserve	Total
San Francisco	Main	600	35	600	17	26	12	20%	1290
Denver	Regional	210	7	210	10	4	0	20%	441
Houston	Regional	155	5	155	10	4	0	20%	329
Remote Office 1	Remote	12	1	12	2	1	0	10%	28
Remote Office 2	Remote	15	1	15	3	1	0	10%	35
Remote Office 3	Remote	8	1	8	3	1	0	10%	21
Total		1000	50	1000	45	37	12		2144

Some additional addresses should be reserved to allow for seamless potential network growth. The commonly suggested reserve is 20 percent for main and regional offices, and 10 percent for remote offices; however, this can vary from case to case. The designer should carefully discuss future network growth with the organization's representative to obtain a more precise estimate of the required resources.

Methods of Assigning IP Addresses

This section discusses methods of assigning IP addresses to end systems and explains their influence on administrative overhead. Address assignment includes assigning an IP address, a default gateway, one or more domain name servers that resolve names to IP addresses, time servers, and so forth. Before selecting the desired IP address assignment method, the following questions should be answered:

- How many devices need an IP address?
- Which devices require static IP address assignment?
- Is IP address renumbering expected in the future?
- Is the administrator required to track devices and their IP addresses?
- Do additional parameters (default gateway, name server, and so forth) have to be configured?
- Are there any availability issues?
- Are there any security issues?

Static Versus Dynamic IP Address Assignment Methods

Following are the two basic IP address assignment strategies:

- **Static:** An IP address is statically assigned to a system. The network administrator configures the IP address, default gateway, and name servers manually by entering them into a special file or files on the end system with either a graphical or text interface. Static address assignment is an extra burden for the administrator—especially on large-scale networks—who must configure the address on every end system in the network.
- **Dynamic:** IP addresses are dynamically assigned to the end systems. Dynamic address assignment relieves the administrator of manually assigning an address to every network device. Instead, the administrator must set up a server to assign the addresses. On that server, the administrator defines the address pools and additional parameters that should be sent to the host (default gateway, name servers, time servers, and so forth). On the host, the

administrator enables the host to acquire the address dynamically; this is often the default. When IP address reconfiguration is needed, the administrator reconfigures the server, which then performs the host-renumbering task. Examples of available address assignment protocols include Reverse Address Resolution Protocol, Boot Protocol, and DHCP. DHCP is the newest and provides the most features.

When to Use Static or Dynamic Address Assignment

To select either a static or dynamic end system IP address assignment method or a combination of the two, consider the following:

- **Node type:** Network devices such as routers and switches typically have static addresses. End-user devices such as PCs typically have dynamic addresses.
- **The number of end systems:** If there are more than 30 end systems, dynamic address assignment is preferred. Static assignment can be used for smaller networks.
- **Renumbering:** If renumbering is likely to happen and there are many end systems, dynamic address assignment is the best choice. With DHCP, only DHCP server reconfiguration is needed; with static assignment, all hosts must be reconfigured.
- **Address tracking:** If the network policy requires address tracking, the static address assignment method might be easier to implement than the dynamic address assignment method. However, address tracking is also possible with dynamic address assignment with additional DHCP server configuration.
- **Additional parameters:** DHCP is the easiest solution when additional parameters must be configured. The parameters have to be entered only on the DHCP server, which then sends the address and those parameters to the clients.
- **High availability:** Statically assigned IP addresses are always available. Dynamically assigned IP addresses must be acquired from the server; if the server fails, the addresses cannot be acquired. To ensure reliability, a redundant DHCP server is required.
- **Security:** With dynamic IP address assignment, anyone who connects to the network can acquire a valid IP address, in most cases. This might be a security risk. Static IP address assignment poses only a minor security risk.

The use of one address assignment method does not exclude the use of another in a different part of the network.

Using DHCP to Assign IP Addresses

DHCP is used to provide dynamic IP address allocation to hosts. DHCP uses a client/server model; the DHCP server can be a Windows server, a UNIX-based server, or a Cisco IOS device. Figure 6-15 shows the steps that occur when a DHCP client requests an IP address from a DHCP server.

Step 1 The host sends a DHCPDISCOVER broadcast message to locate a DHCP server.

Step 2 A DHCP server offers configuration parameters such as an IP address, a MAC address, a domain name, a default gateway, and a lease for the IP address to the client in a DHCPOFFER unicast message.

Step 3 The client returns a formal request for the offered IP address to the DHCP server in a DHCPREQUEST broadcast message.

Step 4 The DHCP server confirms that the IP address has been allocated to the client by returning a DHCPACK unicast message to the client.

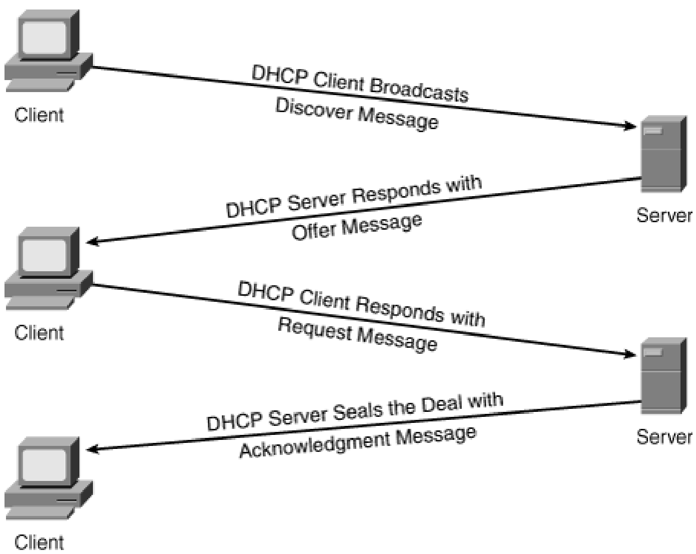

Figure 6-15: DHCP Operation

DCHP Relay

A DHCP relay agent is required to forward DCHP messages between clients and servers when they are on different broadcast domains (IP subnets). For example, the DHCP relay agent receives the DHCPDISCOVER message, which is sent as a broadcast, and forwards it to the DHCP server, on another subnet.

A DHCP client might receive offers from multiple DHCP servers and can accept any one of the offers; the client usually accepts the first offer it receives. An offer from the DHCP server is not a guarantee that the IP address will be allocated to the client; however, the server usually reserves the address until the client has had a chance to formally accept the address.

DHCP supports three possible address allocation mechanisms:

- **Manual:** The network administrator assigns an IP address to a specific MAC address. DHCP is used to dispatch the assigned address to the host.
- **Automatic:** DHCP permanently assigns the IP address to a host.
- **Dynamic:** DHCP assigns the IP address to a host for a limited time (called a *lease*) or until the host explicitly releases the address. This mechanism supports automatic address reuse when the host to which the address has been assigned no longer needs the address.

Selecting Routing Protocol

Types of Routing Protocols

Routing protocols can be classified according to various characteristics. This section gives an overview of the most common IP routing protocols.

Classifying Routing Protocols

Routing protocols can be classified into different groups according to their characteristics. Specifically, routing protocols can be classified by their:

- **Purpose**: Interior Gateway Protocol (IGP) or Exterior Gateway Protocol (EGP)
- **Operation**: Distance vector protocol, link-state protocol, or path-vector protocol
- **Behavior**: Classful (legacy) or classless protocol

For example, IPv4 routing protocols are classified as follows:

- **RIPv1 (legacy)**: IGP, distance vector, classful protocol
- **IGRP (legacy)**: IGP, distance vector, classful protocol developed by Cisco (deprecated from 12.2 IOS and later)
- **RIPv2**: IGP, distance vector, classless protocol
- **EIGRP**: IGP, distance vector, classless protocol developed by Cisco
- **OSPF**: IGP, link-state, classless protocol
- **IS-IS**: IGP, link-state, classless protocol
- **BGP**: EGP, path-vector, classless protocol

The ***classful routing protocols***, RIPv1 and IGRP, are legacy protocols and are only used in older networks. These routing protocols have evolved into the ***classless routing protocols***, RIPv2 and EIGRP, respectively. Link-state routing protocols are classless by nature.

Figure 3-9 displays a hierarchical view of dynamic routing protocol classification.

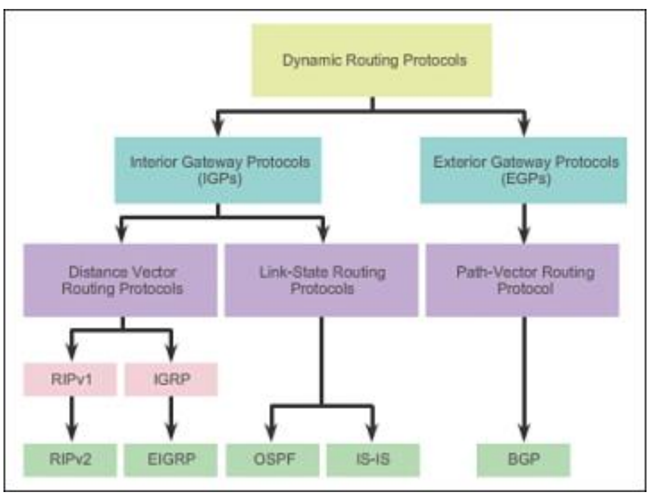

Figure 3-9 Routing Protocol Classification

IGP and EGP Routing Protocols

An ***autonomous system (AS)*** is a collection of routers under a common administration such as a company or an organization. An AS is also known as a routing domain. Typical examples of an AS are a company's internal network and an ISP's network.

The Internet is based on the AS concept; therefore, two types of routing protocols are required:

- ***Interior Gateway Protocols (IGP)***: Used for routing within an AS. It is also referred to as intra-AS routing. Companies, organizations, and even service providers use an IGP on their internal networks. IGPs include RIP, EIGRP, OSPF, and IS-IS.
- ***Exterior Gateway Protocols (EGP)***: Used for routing between autonomous systems. It is also referred to as inter-AS routing. Service providers and large companies may interconnect using an EGP. The Border Gateway Protocol (BGP) is the only currently viable EGP and is the official routing protocol used by the Internet.

NOTE

Because BGP is the only EGP available, the term EGP is rarely used; instead, most engineers simply refer to BGP.

The example in Figure 3-10 provides simple scenarios highlighting the deployment of IGPs, BGP, and static routing.

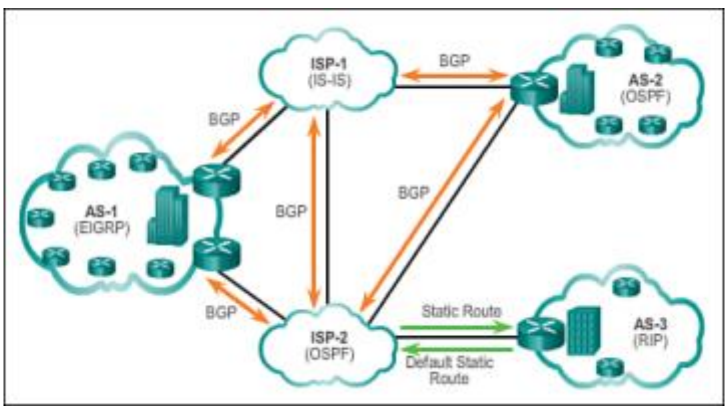

Figure 3-10 IGP versus EGP Routing Protocols

There are five individual autonomous systems in the scenario:

- **ISP-1**: This is an AS and it uses IS-IS as the IGP. It interconnects with other autonomous systems and service providers using BGP to explicitly control how traffic is routed.
- **ISP-2**: This is an AS and it uses OSPF as the IGP. It interconnects with other autonomous systems and service providers using BGP to explicitly control how traffic is routed.
- **AS-1**: This is a large organization and it uses EIGRP as the IGP. Because it is multihomed (i.e., connects to two different service providers), it uses BGP to explicitly control how traffic enters and leaves the AS.

- **AS-2**: This is a medium-sized organization and it uses OSPF as the IGP. It is also multihomed; therefore, it uses BGP to explicitly control how traffic enters and leaves the AS.
- **AS-3**: This is a small organization with older routers within the AS; it uses RIP as the IGP. BGP is not required because it is single-homed (i.e., connects to one service provider). Instead, static routing is implemented between the AS and the service provider.

Route summarization

Route summarization means summarizing a group of routes into a single route advertisement. The net result of route summarization and its most obvious benefit is a reduction is the size of routing tables on the network. This in turn reduces the latency associated with each router hop since the average speed for routing table lookup will be increased due to the reduced number of entries. The routing protocol overhead can also be significantly reduced since fewer routing entries are being advertised. This can become critical as the overall network (and hence the number of subnets) grows.

In order to implement route summarization that can be arbitrarily configured a classless routing protocol is required, however that in itself is not enough. It is imperative to plan the IP addressing scheme such that non-conflicting summarization can be performed at strategic points in the network. These ranges are called contiguous address blocks. For example a router that connects a group of branch offices to the head office could summarize all of the subnets used by those branch offices into a single route advertisement. If the subnets all fell within the range 172.16.16.0/24 to 172.16.31.0/24 then the range could be summarized as 172.16.16.0/20. This is a contiguous range that also coincides with a perfect bit boundary thus ensuring that the address range can be summarized in a single statement. Clearly to maximize the benefits of route summarization careful address planning is essential.

BGP

The Border Gateway Protocol (BGP) routes traffic between autonomous systems. An autonomous system is a network or group of networks under common administration and with common routing policies. BGP exchanges routing information for the Internet and is the protocol used between ISPs. Customer networks, such as universities and corporations, usually employ an Interior Gateway Protocol (IGP), such as RIP or OSPF, to exchange routing information within their networks. Customers connect to ISPs, and ISPs use BGP to exchange customer and ISP routes. When BGP is used between autonomous systems, the protocol is referred to as external BGP (eBGP). If a service provider is using BGP to exchange routes within an autonomous system, the protocol is referred to as interior BGP (iBGP).

BGP is a very robust and scalable routing protocol, as evidenced by the fact that it is the routing protocol employed on the Internet. To achieve scalability at this level, BGP uses many route parameters, called attributes, to define routing policies and maintain a stable routing environment. BGP neighbors exchange full routing information when the TCP connection between neighbors is first established. When changes to the routing table are detected, the BGP routers send to their neighbors only those routes that have changed. BGP routers do not send periodic routing updates, and BGP routing updates advertise only the optimal path to a destination network.

MP-BGP

Multiprotocol BGP (MP-BGP) adds capabilities to BGP to enable multicast routing policy throughout the Internet and to connect multicast topologies within and between BGP autonomous systems. That is, MP-BGP is an enhanced BGP that carries IP multicast routes. BGP carries two sets of routes, one set for unicast routing and one set for multicast routing. The routes associated with multicast routing are used by the Protocol Independent Multicast (PIM) to build data distribution trees.

eBGP/iBGP

As noted previously, BGP is an interautonomous system routing protocol. When BGP is used between autonomous systems (AS), the protocol is referred to as external BGP (eBGP). If a service provider is using BGP to exchange routes within an AS, then the protocol is referred to as interior BGP (iBGP).

OSPF

Open Shortest Path First (OSPF) is a routing protocol developed for IP networks by the IGP working group of the Internet Engineering Task Force (IETF). It was derived from several research efforts, including a version of OSI's IS-IS routing protocol.

OSPF has two primary characteristics:

- It is an open protocol. Its specification is in the public domain (RFC 1247).
- It is based on the Shortest Path First (SPF) algorithm, sometimes known as the Dijkstra algorithm.

OSPF is a link-state routing protocol that calls for the sending of link-state advertisements (LSAs) to all other routers within the same hierarchical area. Information on attached interfaces, metrics used, and other variables are included in OSPF LSAs. As OSPF routers accumulate link-state information, they use the SPF algorithm to calculate the shortest path to each node.

EIGRP

Enhanced Interior Gateway Routing Protocol (EIGRP) is Cisco's proprietary routing protocol, based on IGRP. EIGRP is a distance-vector routing protocol, with optimizations to minimize routing instability incurred after topology changes, and the use of bandwidth and processing power in the router. Routers that support EIGRP will automatically redistribute route information to IGRP neighbors by converting the 32-bit EIGRP metric to the 24-bit IGRP metric. Most of the routing optimizations are based on the Diffusing Update Algorithm (DUAL), which guarantees loop-free operation and provides fast router convergence.

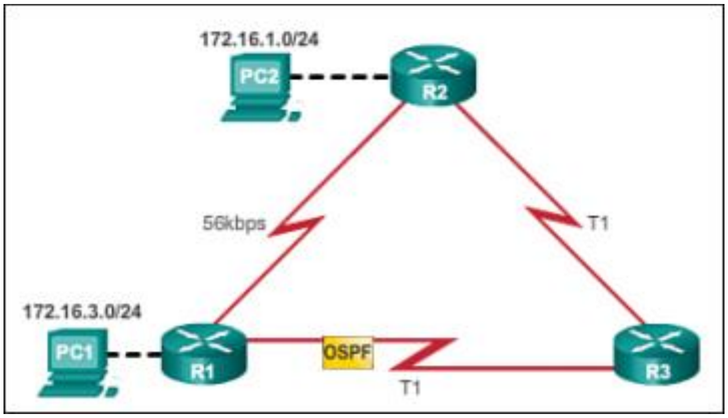

Figure 3-20 OSPF Uses Faster Links

RIP

The Routing Information Protocol (RIP) is one of the oldest routing protocols still in wide use. Today's open standard version of RIP, sometimes referred to as IP RIP, is formally defined in RFC 1058 and in STD 56. RIP is a distance-vector routing protocol that uses hop count as a metric. RIP prevents routing loops by implementing a limit on the number of hops allowed in source/destination paths, and also implements split horizon, route poisoning and holddown mechanisms to prevent incorrect routing information from being propagated.

For example, assume that PC1 wants to send a packet to PC2. In Figure 3-19, the RIP routing protocol has been enabled on all routers and the network has converged. RIP makes a routing protocol decision based on the least number of hops. Therefore, when the packet arrives on R1, the best route to reach the PC2 network would be to send it directly to R2 even though the link is much slower that all other links.

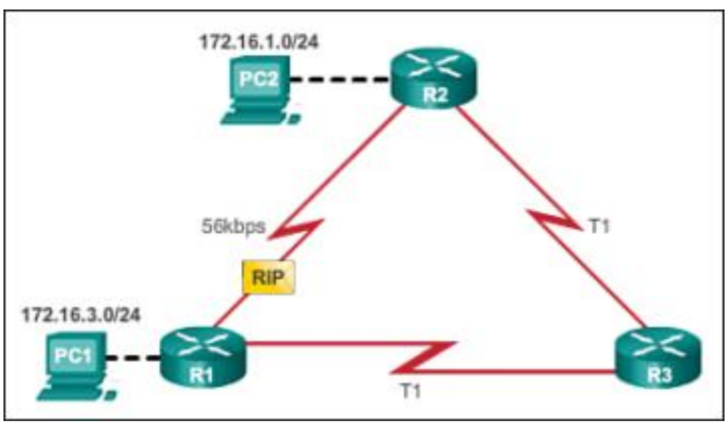

Figure 3-19 RIP Uses Shortest Hop Count Path

IS-IS

Intermediate system to intermediate system (IS-IS) is a link-state routing protocol. It operates by reliably flooding topology information throughout a network of routers. Each router then builds its own picture of the network's topology. Packets or datagrams are forwarded based on the best topological path through the network; IS-IS uses Dijkstra's algorithm for computing best paths.

IS-IS was first defined in ISO/IEC 10589:2002 and was republished in RFC 1142 for the Internet community. IS-IS is an IGP, intended for use within one administrative domain or network only.

Choosing the routing protocol

Routing protocols can be compared based on the following characteristics:

- **Speed of convergence**: Speed of convergence defines how quickly the routers in the network topology share routing information and reach a state of consistent knowledge. The faster the convergence, the more preferable the protocol. Routing loops can occur when inconsistent routing tables are not updated due to slow convergence in a changing network.
- **Scalability**: Scalability defines how large a network can become, based on the routing protocol that is deployed. The larger the network is, the more scalable the routing protocol needs to be.
- **Classful or classless (use of VLSM)**: Classful routing protocols do not include the subnet mask and cannot support ***variable-length subnet mask (VLSM)***. Classless routing protocols include the subnet mask in the updates. Classless routing protocols support VLSM and better route summarization.
- **Resource usage**: Resource usage includes the requirements of a routing protocol such as memory space (RAM), CPU utilization, and link bandwidth utilization. Higher resource requirements necessitate more powerful hardware to support the routing protocol operation, in addition to the packet forwarding processes.
- **Implementation and maintenance**: Implementation and maintenance describes the level of knowledge that is required for a network administrator to implement and maintain the network based on the routing protocol deployed.

	Distance Vector				Link State	
	RIPv1	RIPv2	IGRP	EIGRP	OSPF	IS-IS
Speed Convergence	Slow	Slow	Slow	Fast	Fast	Fast
Scalability - Size of Network	Small	Small	Small	Large	Large	Large
Use of VLSM	No	Yes	No	Yes	Yes	Yes
Resource Usage	Low	Low	Low	Medium	High	High
Implemenation and Maintenance	Simple	Simple	Simple	Complex	Complex	Complex